

GHOSTFACE — Export-Control Classification: Technical Memorandum for Counsel

From: Benjamin Henderson, Director, Ghostface Limited (NZ) **Re:** Encryption classification for App Store / Play Store distribution **Date:** 19 August 2026 **Status:** Company’s technical understanding, prepared to focus counsel’s review. **Not legal advice and not a self-classification decision** — we are seeking counsel’s opinion on the questions in §5.

1. What GHOSTFACE is

GHOSTFACE is a free, publicly-available end-to-end encrypted messaging and calling app for iOS and Android, published by Ghostface Limited, a New Zealand company. Its cryptography serves one purpose: protecting users’ personal communications in transit and at rest. See the accompanying **Cryptographic Inventory** for the full technical detail.

2. Why we are seeking advice

The app is distributed through Apple (US-based infrastructure) and is built on Expo’s US-based build servers, so US Export Administration Regulations (EAR) exposure exists regardless of the company’s New Zealand domicile. As a New Zealand exporter, we may also have obligations under New Zealand’s strategic-goods controls (administered by MFAT). We want a defensible classification under both regimes **before** resuming distribution. Two builds were previously distributed via TestFlight, with Apple’s export questionnaire answered, before we identified the specific question below; we then paused.

3. The cryptography, in brief

Every algorithm used is a standard published algorithm: ChaCha20-Poly1305 (RFC 8439), X25519 (RFC 7748), Ed25519 (RFC 8032), ML-KEM-768 (NIST FIPS 203), SHA-2 (FIPS 180-4), HKDF (RFC 5869), PBKDF2 (NIST SP 800-132), all via the open-source @noble/* libraries, over the OS-provided CSPRNG. The protocol is the publicly-documented Signal design (X3DH + Double Ratchet), extended to post-quantum using the same hybrid method Signal published as PQXDH and Apple published as iMessage PQ3.

4. The specific point we want tested

We believe GHOSTFACE is a **mass-market** product using **only standard cryptography**, which — as we understand it — would point toward mass-market treatment (US: ECCN 5D992.c self-classification under EAR §740.17; NZ: the Wassenaar “Cryptography Note” / mass-market exemption) rather than a formal review (US: a CCATS filing with BIS; NZ: an MFAT permit).

The one element that we want counsel to scrutinise is an internally-named function, **kdfRkPQ**. We flag it proactively because the name might suggest a proprietary algorithm. It is not. It is a single HKDF-HMAC-SHA256 (RFC 5869) invocation whose input is the concatenation of a standard X25519 Diffie-Hellman output and a standard ML-KEM-768 shared secret, with a domain-separation label. It introduces no new cryptographic primitive; it is the standard, published technique for combining a classical and a post-quantum secret, and is structurally the same as the continuous post-quantum rekeying in Apple’s iMessage PQ3.

Our understanding is therefore that this does **not** constitute “non-standard” or “proprietary” cryptography in the sense that would force a formal classification, but we recognise this is a regulatory-interpretation question, not a technical one — which is why we are asking rather than self-classifying.

5. Questions for counsel

1. **US EAR:** Does GHOSTFACE qualify for mass-market self-classification (5D992.c under §740.17), or does any element — in particular the `kdfRkPQ` composition — require a CCATS classification request to BIS? Are there notification/self-classification report obligations we must meet either way?
2. **New Zealand:** Does the Wassenaar Cryptography Note / mass-market exemption cover export of this software from New Zealand, or is an MFAT strategic-goods permit required?
3. **Practically:** What is the lowest-friction compliant path to resume App Store / Play Store distribution, and are there records we should retain (e.g. the classification rationale, this inventory) to evidence compliance?

6. What we can provide

The full cryptographic inventory (attached), the relevant source files, the Apple export-questionnaire responses from the prior builds, and any additional technical detail on request. The author can answer technical questions on the crypto stack directly.

Prepared by the company to assist counsel’s assessment. Ghostface Limited is not qualified to make the export-control determination and is engaging counsel precisely to obtain it.